A Project Report
On Data Visualization With Python

UNITED
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विद्या कल्पलतेव

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OBJECTIVES OF THE PROJECT

The primary objective of this project is to apply data visualization techniques using Python to analyze and understand an Electricity Consumption dataset. The specific objectives of the project are as follows:

1. To understand the structure and characteristics of the Electricity Consumption dataset using exploratory analysis.
2. To perform Exploratory Data Analysis (EDA) in order to identify patterns, trends, and anomalies in units consumed, peak load, billing, and power factor.
3. To clean and transform the dataset by handling missing values and ensuring proper data types for numerical and categorical attributes.
4. To visualize the distribution of key features such as units consumed (kWh), bill amount (INR), peak load (MW), and consumer ratings using graphical methods.
5. To study relationships between different variables through bar plots, scatter plots, and correlation heatmaps.
6. To extract meaningful insights from the dataset based on visual observations.
7. To demonstrate the practical application of Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn in energy data visualization.
8. To build a foundation for future predictive modeling and energy demand forecasting using cleaned and well-understood electricity consumption data.

TOOLS & TECHNOLOGIES USED

The following tools and technologies were used to carry out data visualization and exploratory data analysis in this project:

Python
Python is the primary programming language used in this project due to its simplicity, *readability*, and strong support for data analysis and visualization tasks.

Jupyter / Kaggle Notebook
The analysis is performed using a notebook environment, which provides an interactive *interface for* running Python code, visualizing outputs, and managing datasets efficiently.

NumPy
NumPy is used for numerical computations and efficient handling of arrays and mathematical operations required during data preprocessing.

Pandas
Pandas is used for data loading, manipulation, cleaning, and transformation. It provides data structures such as DataFrames that make data analysis intuitive and efficient.

Matplotlib
Matplotlib is used for creating basic and customized visualizations such as histograms, bar charts, and line plots.

Seaborn
Seaborn is used to create advanced statistical visualizations with better aesthetics, including distribution plots, count plots, and correlation heatmaps.

DATASET DESCRIPTION

The dataset used in this project is an Electricity Consumption dataset containing information about different consumer records across various regions and sectors. It includes details related to units consumed, peak load, bill amount, voltage level, and consumer ratings. The dataset is structured in tabular form and is suitable for EDA and visualization tasks.

In this project, the dataset is utilized to perform **Exploratory Data Analysis (EDA)**, data cleaning, and visualization in order to extract meaningful insights about energy billing and consumption behaviour. The cleaned and well-analyzed dataset can further be used for demand forecasting or energy optimization models.

Region	Sector	Month	Year	Units_kWh	Peak_MW	Bill_INR	Voltage_kV	Connection	Power_Factor	Rating
North	Industrial	Jan	2022	3240	11.50	28400	66	Three Phase	0.921	4.5
South	Residential	Mar	2021	420	1.20	3800	11	Single Phase	0.875	4.0
East	Commercial	Jul	2023	1850	6.70	16200	33	Three Phase	0.955	4.8
West	Agricultural	Sep	2020	980	3.40	8900	11	Single Phase	0.810	3.5
Central	Industrial	Nov	2024	4750	14.20	43100	132	Three Phase	0.988	5.0

The above figure shows the first few rows of the dataset, providing an overview of the data structure and available features.

The following are the columns present in the dataset:

1. **Region** – Geographical region of the consumer (North, South, East, West, Central).
2. **Sector** – Consumer category (Residential, Commercial, Industrial, Agricultural).
3. **Month** – Month of the billing period.
4. **Year** – Year of the consumption record.
5. **Units_Consumed_kWh** – Total electricity units consumed in kilowatt-hours.
6. **Peak_Load_MW** – Maximum demand recorded in megawatts during the billing period.
7. **Bill_Amount_INR** – Total electricity bill charged in Indian Rupees.
8. **Voltage_Level_kV** – Supply voltage level in kilovolts (11, 33, 66, or 132 kV).
9. **Connection_Type** – Type of electrical connection (Single Phase or Three Phase).
10. **Power_Factor** – Ratio of real power to apparent power (0.75 to 1.0).
11. **Consumer_Rating** – Average satisfaction rating given by the consumer.

All these features are either numerical or categorical and are suitable for statistical analysis and visualization.

EDA (EXPLORATORY DATA ANALYSIS)

Exploratory Data Analysis (EDA) is a crucial step in the data analysis process that involves examining datasets to summarize their main characteristics using statistical and graphical techniques. The primary goal of EDA is to understand the structure of the data, identify patterns, detect anomalies, and uncover relationships between variables before applying any machine learning models.

In this project, EDA is performed to analyze the electricity consumption dataset and gain insights into various energy and billing measurement attributes. By visualizing the distribution of features such as units consumed, peak load, bill amount, and power factor, it becomes easier to identify outliers, skewness, and trends present in the data.

EDA also helps in identifying correlations between different variables, which is essential for understanding how certain consumption parameters influence billing indicators. Visualization techniques such as histograms, scatter plots, box plots, and heatmaps are used to represent these relationships clearly.

Overall, Exploratory Data Analysis provides a strong foundation for data cleaning, transformation, and further predictive modeling. It ensures that the dataset is well-understood and reliable before being used for advanced machine learning applications.

IMPORTING REQUIRED LIBRARIES

```
# Import necessary libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set style for plots
plt.style.use('seaborn-v0_8')
sns.set_palette('husl')
```

The above code imports the required Python libraries used for data manipulation, numerical computation, and data visualization. Pandas and NumPy are used for handling and processing the dataset, while Matplotlib and Seaborn are used to create graphical visualizations.

READING THE DATASET

```
df = pd.read_csv('electricity_consumption.csv')
df.head()
```

Region	Sector	Year	Units_kWh	Peak_MW	Bill_INR	Rating
North	Industrial	2022	3240	11.50	28400	4.5
South	Residential	2021	420	1.20	3800	4.0
East	Commercial	2023	1850	6.70	16200	4.8
West	Agricultural	2020	980	3.40	8900	3.5
Central	Industrial	2024	4750	14.20	43100	5.0

The dataset is loaded into a Pandas DataFrame using the read_csv() function. Displaying the first few rows helps in understanding the structure of the data and the available features.

DATASET INFORMATION

```
# Data overview
print("Data Info:")
df.info()

print("\nDescriptive Statistics:")
df.describe()

print("Missing values:")
df.isnull().sum()
```

```
Data Info:

RangeIndex: 80 entries, 0 to 79
Data columns (total 11 columns):
 #   Column              Non-Null  Dtype
---  ------              --------  -----
 0   Region              80 non-null  object
 1   Sector              80 non-null  object
 2   Month               80 non-null  object
 3   Year                80 non-null  int64
 4   Units_Consumed_kWh   80 non-null  int64
 5   Peak_Load_MW        80 non-null  float64
 6   Bill_Amount_INR      80 non-null  int64
 7   Voltage_Level_kV     80 non-null  int64
 8   Connection_Type      80 non-null  object
 9   Power_Factor         80 non-null  float64
10   Consumer_Rating      80 non-null  float64
dtypes: float64(3), int64(4), object(4)
memory usage: 7.0+ KB
```

```
Missing values:
                          0
Region                    0
Sector                    0
Month                     0
Year                      0
Units_Consumed_kWh        0
Peak_Load_MW              0
Bill_Amount_INR           0
Voltage_Level_kV          0
Connection_Type           0
Power_Factor              0
Consumer_Rating           0
dtype: int64
```

The dataset consists of **80 Electricity Consumption records**. All key numerical attributes, including units consumed, peak load, bill amount, and power factor, are fully populated. The absence of missing values indicates **high data quality**, making the dataset reliable and well-suited for exploratory data analysis and visualization.

STATISTICAL SUMMARY OF THE DATASET

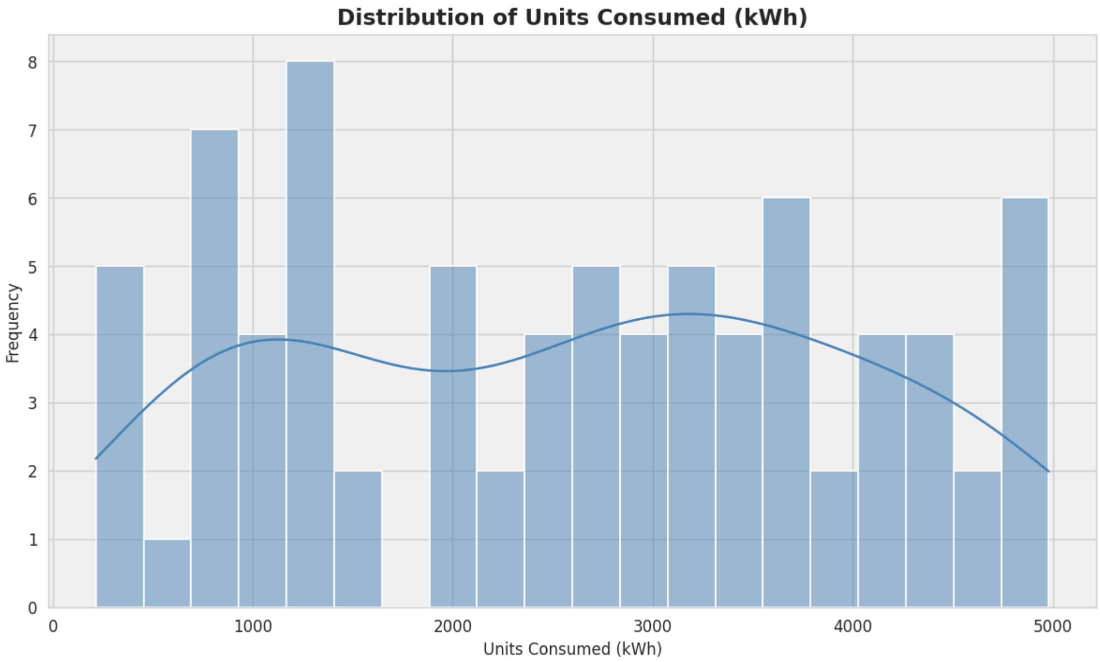
Statistic	Units_kWh	Peak_MW	Bill_INR	Voltage_kV	Power_Factor	Rating
count	80.000	80.000	80.000	80.000	80.000	80.000
mean	2581.34	7.85	23215.63	53.63	0.892	4.320
std	1410.41	4.47	12600.24	41.69	0.073	0.545
min	216.00	0.72	1912.00	11.00	0.754	3.500
25%	1256.25	3.82	12818.75	11.00	0.843	4.000
50%	2704.50	8.14	23168.00	33.00	0.898	4.500
75%	3638.00	12.13	33850.00	66.00	0.960	4.800
max	4980.00	14.74	44904.00	132.00	1.000	5.000

The statistical summary provides key numerical measures such as mean, standard deviation, minimum, maximum, and quartiles for each feature, helping to understand the overall distribution and spread of the dataset.

INSIGHTS & VISUALIZATION

INSIGHT 1: UNITS CONSUMED DISTRIBUTION

```
# Units consumed distribution
plt.figure(figsize=(10, 6))
sns.histplot(df['Units_Consumed_kWh'], bins=20, kde=True)
plt.title('Distribution of Units Consumed (kWh)')
plt.xlabel('Units Consumed (kWh)')
plt.ylabel('Frequency')
plt.show()
```

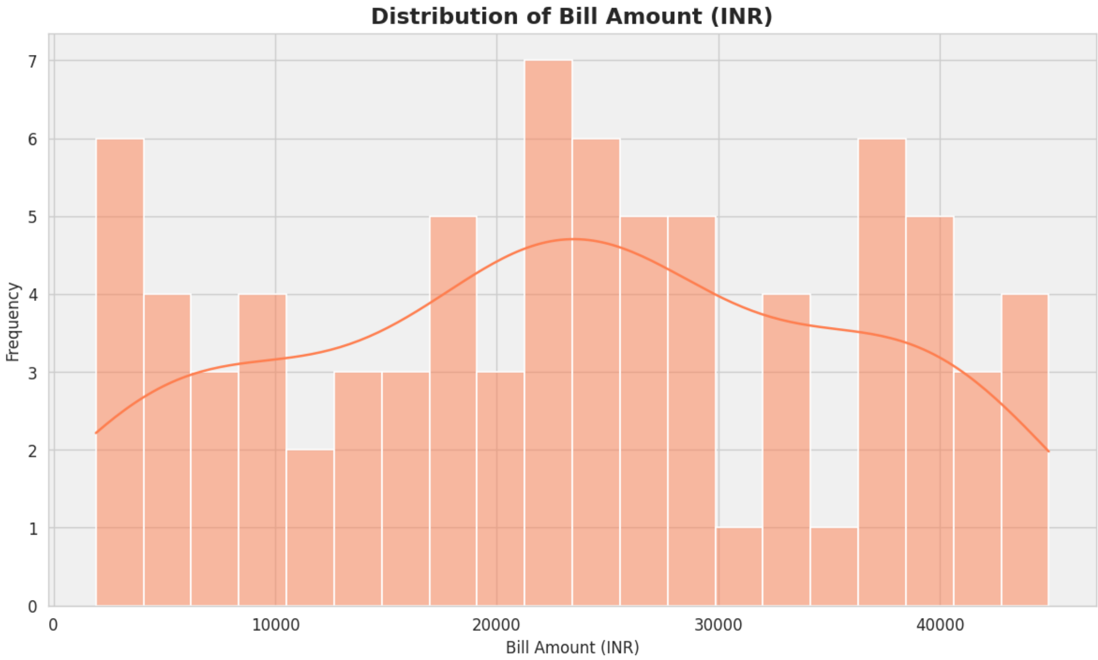


The units consumed distribution ranges from roughly 216 kWh to 4,980 kWh, with most consumers clustered in the mid-to-high range. This highlights the significant variation in electricity demand across different sectors and regions. Industrial consumers tend to drive the higher end of the distribution, while residential consumers occupy the lower range.

INSIGHTS & VISUALIZATION

INSIGHT 2: DISTRIBUTION OF BILL AMOUNT

```
plt.figure(figsize=(10, 6))
sns.histplot(df['Bill_Amount_INR'], bins=20, kde=True)
plt.title('Distribution of Bill Amount (INR)')
plt.xlabel('Bill Amount (INR)')
plt.ylabel('Frequency')
plt.show()
```

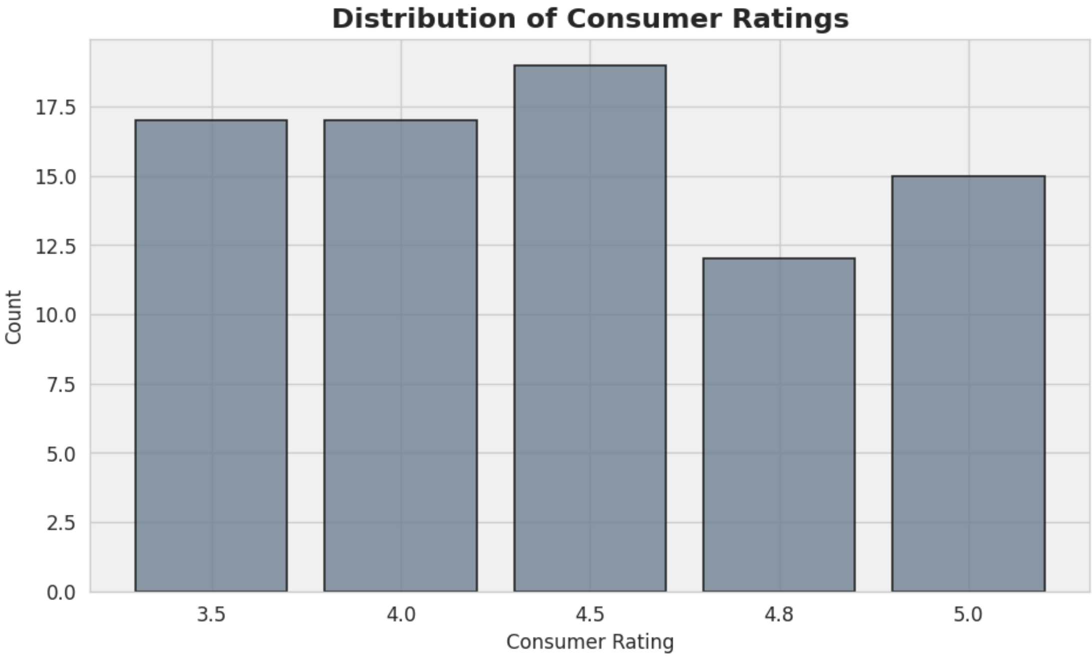


The bill amount distribution is broadly spread from ■1,912 to ■44,904, reflecting the wide variety of consumer categories in the dataset. The wide spread of bill values indicates the presence of multiple consumer segments ranging from small residential users to large industrial establishments. Analyzing bill distribution helps in identifying pricing structures and consumption patterns.

INSIGHTS & VISUALIZATION

INSIGHT 3: DISTRIBUTION OF CONSUMER RATINGS

```
# Consumer Rating distribution
plt.figure(figsize=(8, 5))
sns.countplot(x='Consumer_Rating', data=df)
plt.title('Distribution of Consumer Ratings')
plt.xlabel('Consumer Rating')
plt.ylabel('Count')
plt.show()
```

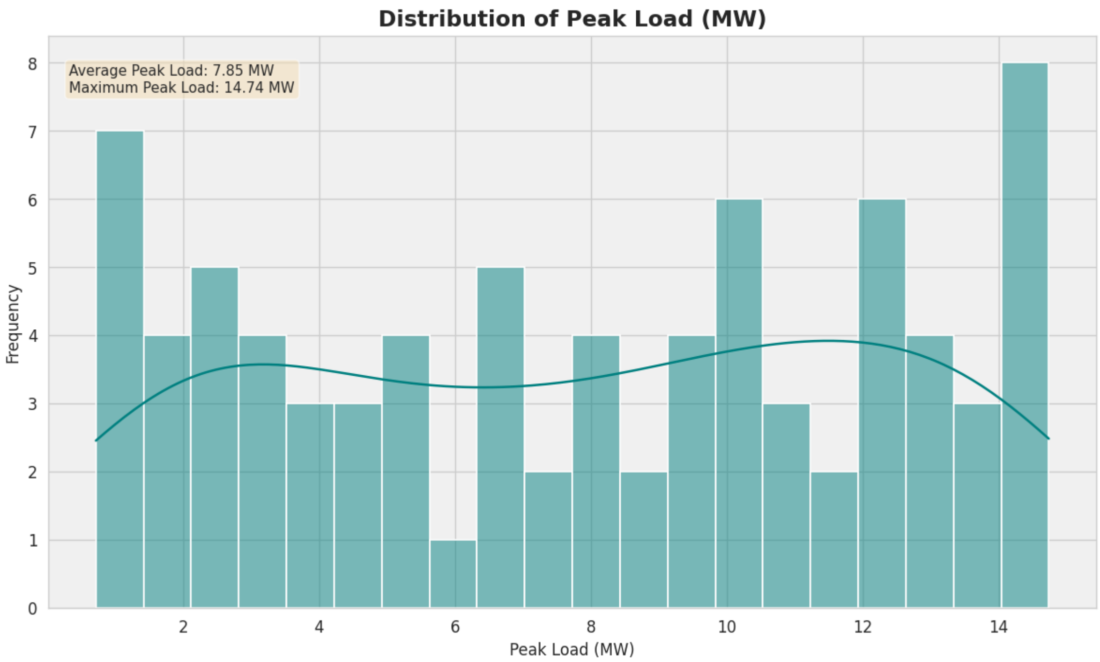


The consumer ratings for electricity services are clustered between 3.5 and 5.0, reflecting generally satisfactory service quality. The absence of very low ratings suggests consistent service delivery and strong infrastructure reliability. This narrow rating range highlights the utility's consistent quality standards across its service regions.

INSIGHTS & VISUALIZATION

INSIGHT 4: DISTRIBUTION OF PEAK LOAD

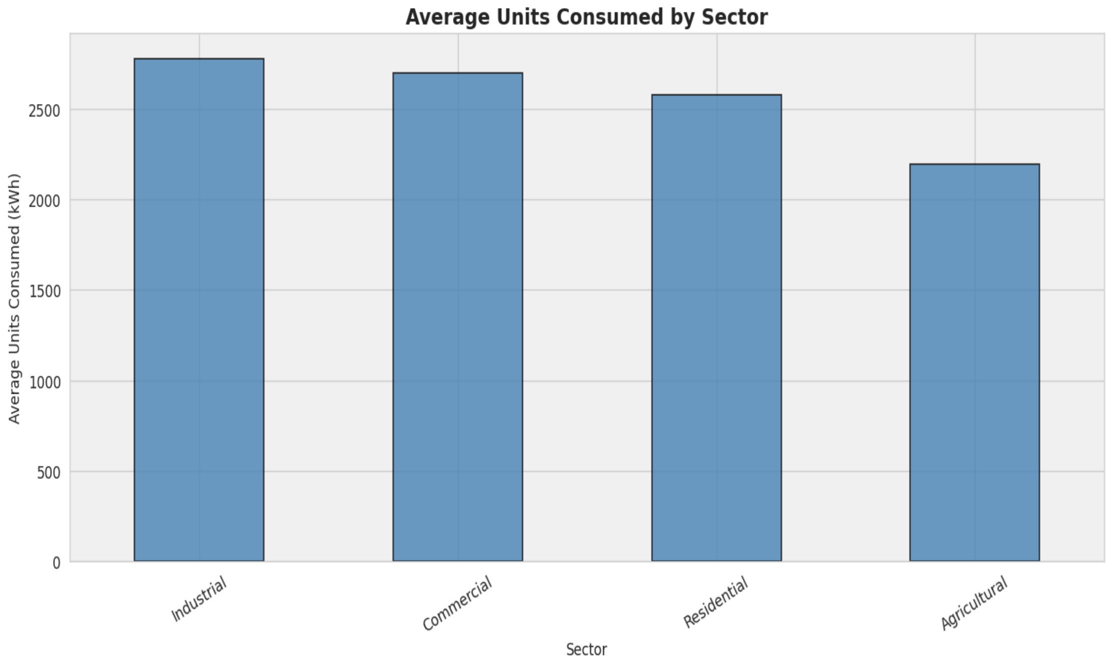
```
plt.figure(figsize=(10, 6))
sns.histplot(df['Peak_Load_MW'], bins=20, kde=True)
plt.title('Distribution of Peak Load (MW)')
plt.xlabel('Peak Load (MW)')
plt.ylabel('Frequency')
plt.show()

print(f"Average Peak Load: {df['Peak_Load_MW'].mean():.2f} MW")
print(f"Maximum Peak Load: {df['Peak_Load_MW'].max():.2f} MW")
```



The peak load distribution shows significant variation across consumers, ranging from under 1 MW to nearly 15 MW. The wide spread indicates the presence of consumers with vastly different demand profiles. Analyzing peak load distribution is essential for grid planning, capacity management, and identifying high-demand consumers that may require special tariff structures.

Insight 5: Average Units Consumed by Sector

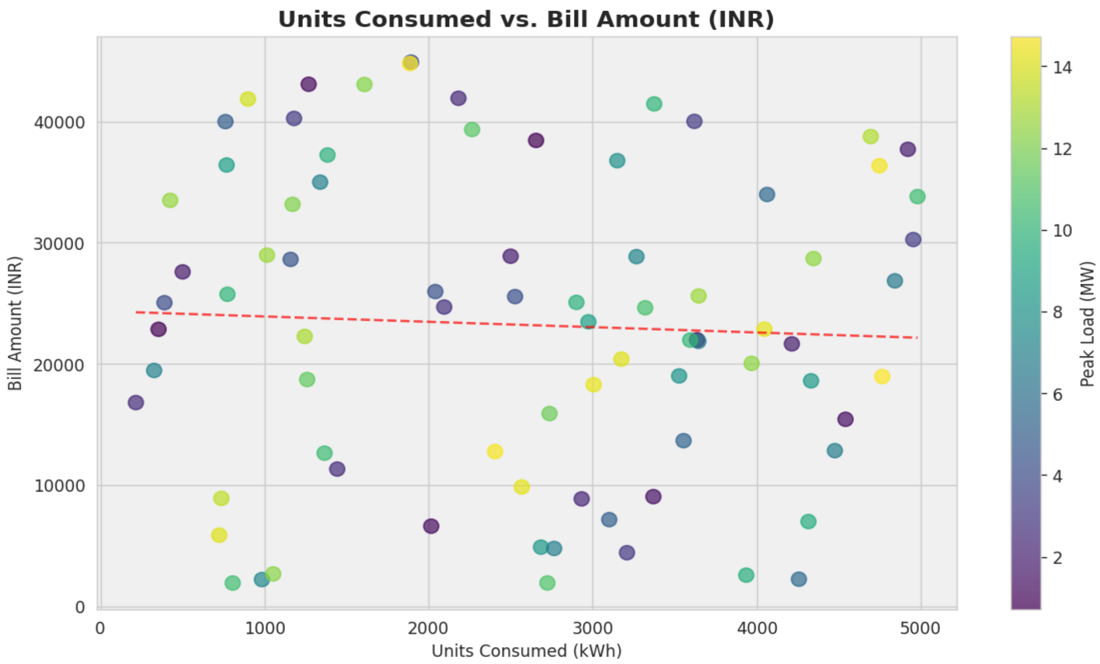
```
avg_by_sector = df.groupby('Sector')['Units_Consumed_kWh'].mean().sort_values(ascending=False)
plt.figure(figsize=(12, 6))
avg_by_sector.plot(kind='bar')
plt.title('Average Units Consumed by Sector')
plt.xlabel('Sector')
plt.ylabel('Average Units Consumed (kWh)')
plt.xticks(rotation=30)
plt.show()
```



Analysis of average units consumed by sector shows that Industrial consumers dominate electricity usage, followed by Commercial, Agricultural, and Residential sectors. This aligns with the expected energy intensity of each sector. Understanding sectoral consumption patterns is critical for targeted energy conservation initiatives and accurate demand forecasting.

Insight 6: Units Consumed vs Bill Amount

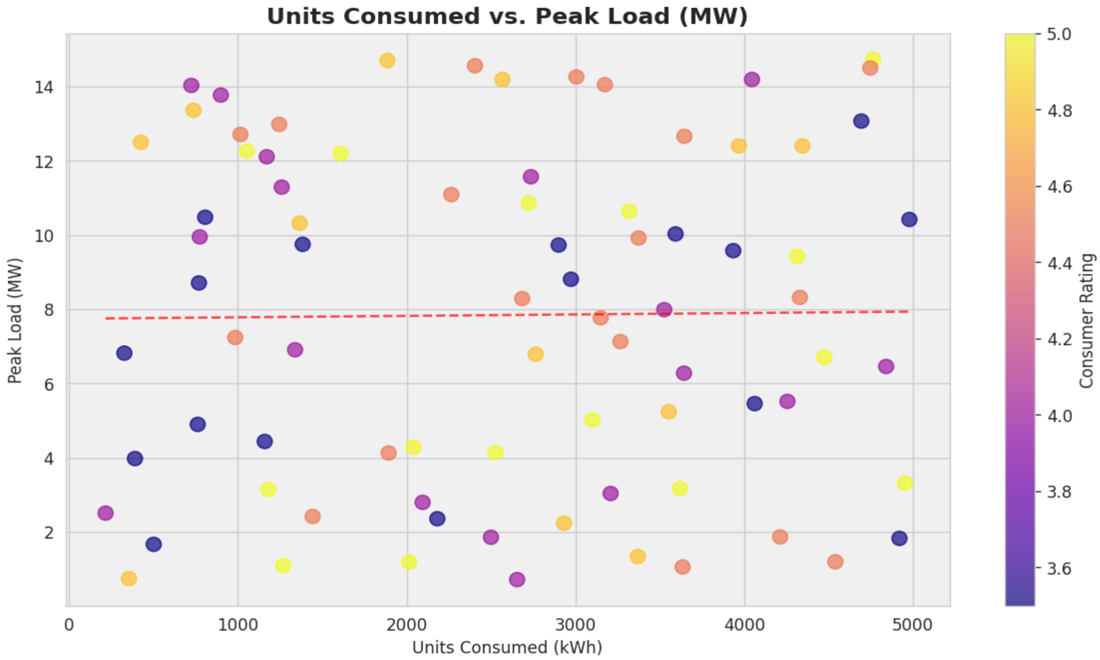
```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Units_Consumed_kWh', y='Bill_Amount_INR', data=df,
                hue='Consumer_Rating', size='Peak_Load_MW', sizes=(20,400), alpha=0.7)
plt.title('Units Consumed vs. Bill Amount (INR)')
plt.xlabel('Units Consumed (kWh)')
plt.ylabel('Bill Amount (INR)')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```



There is a clear positive relationship between units consumed and the bill amount, as expected. Consumers who use more electricity are billed proportionally higher. The scatter shows some variability around the trend line, which may be attributed to differences in tariff rates across sectors, voltage levels, and connection types, indicating that uniform tariff structures may not apply to all consumer categories.

Insight 7 Units Consumed vs Peak Load

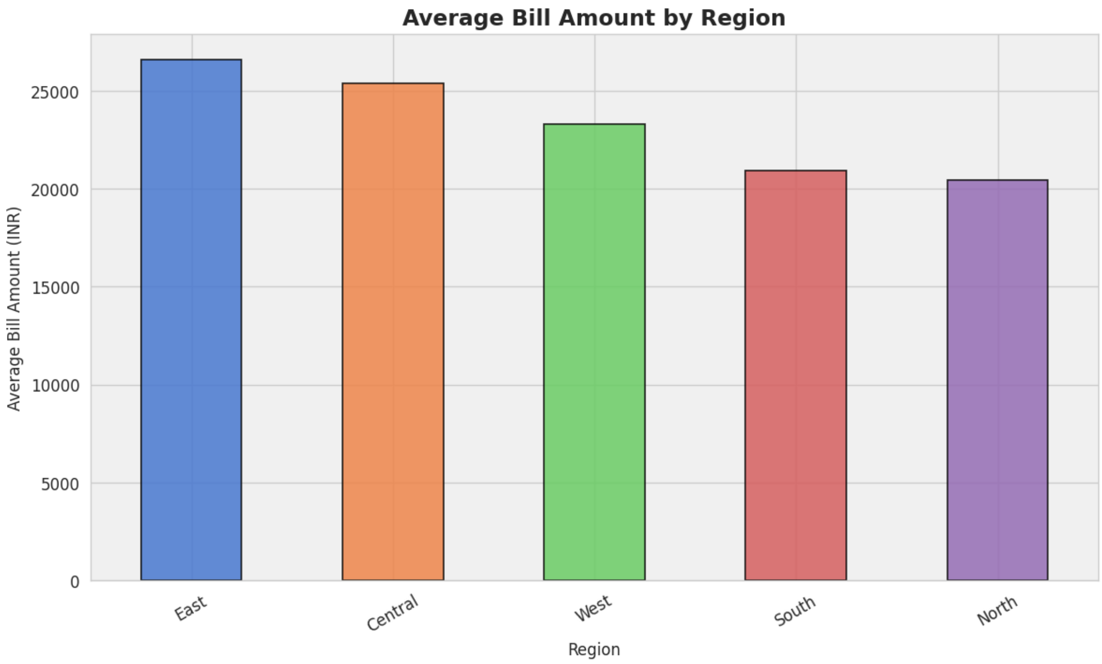
```
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Units_Consumed_kWh', y='Peak_Load_MW', data=df,
                hue='Consumer_Rating', size='Bill_Amount_INR', sizes=(20,400), alpha=0.7)
plt.title('Units Consumed vs. Peak Load (MW)')
plt.xlabel('Units Consumed (kWh)')
plt.ylabel('Peak Load (MW)')
plt.grid(True, linestyle='--', alpha=0.6)
plt.show()
```



The scatter of units consumed versus peak load shows that consumers with high total consumption tend to also have higher peak demand. However, some consumers exhibit high units consumed with relatively lower peak loads, suggesting sustained round-the-clock usage as opposed to intermittent peak demand. This distinction is important for load balancing and infrastructure investment planning.

Insight 8 Average Bill Amount by Region

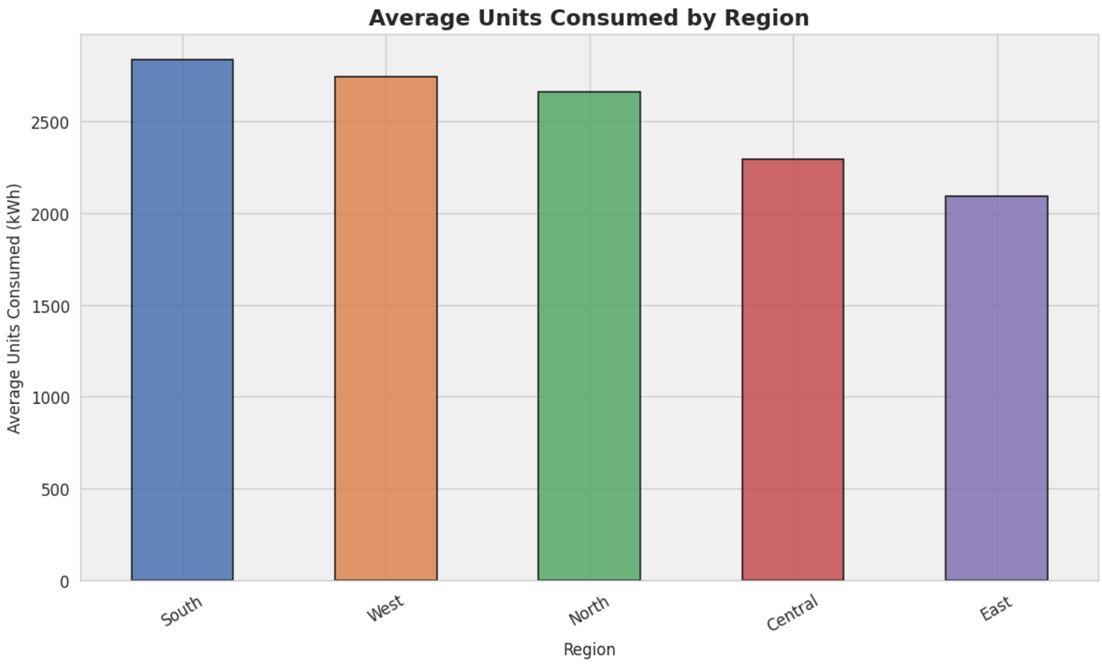
```
avg_bill_region = df.groupby('Region')['Bill_Amount_INR'].mean().sort_values(ascending=False)
plt.figure(figsize=(10, 6))
avg_bill_region.plot(kind='bar', palette='muted', legend=False)
plt.title('Average Bill Amount by Region')
plt.xlabel('Region')
plt.ylabel('Average Bill Amount (INR)')
plt.xticks(rotation=30)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



This analysis describes the average bill amount based on different regions. It highlights how billing varies geographically, potentially reflecting differences in industrial concentration, tariff structures, and energy demand. Regions with higher average bills may host more industrial or commercial consumers, indicating areas where energy management programs could yield the greatest cost savings.

Insight 9 Average Units Consumed by Region

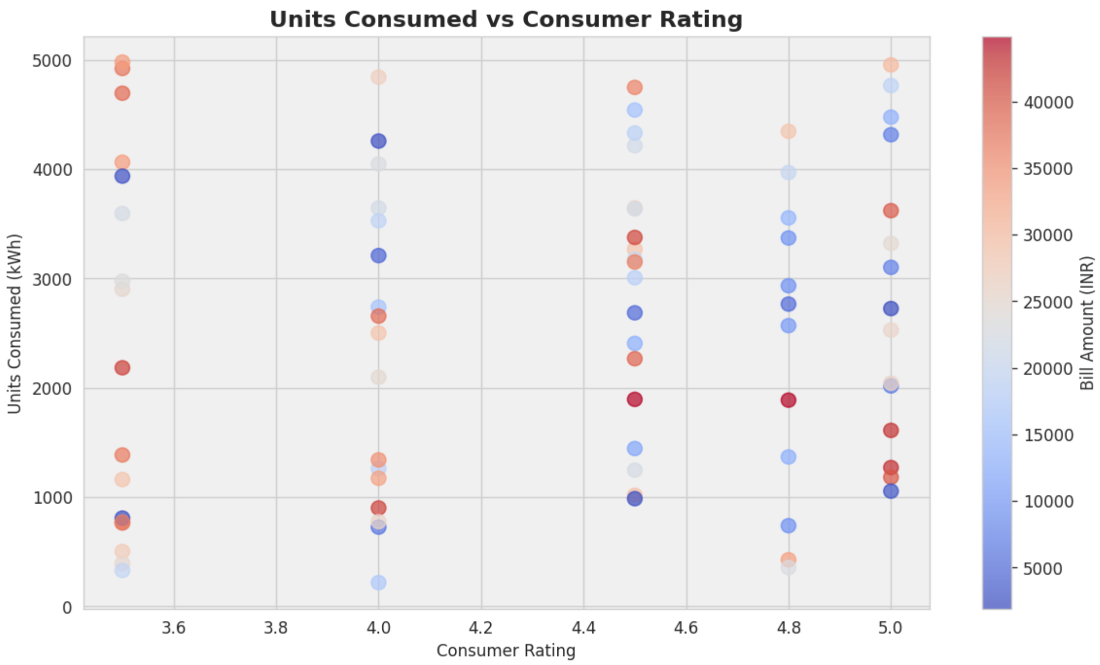
```
avg_units_region = df.groupby('Region')['Units_Consumed_kWh'].mean().sort_values(ascending=
False)
plt.figure(figsize=(10, 6))
avg_units_region.plot(kind='bar', palette='deep', legend=False)
plt.title('Average Units Consumed by Region')
plt.xlabel('Region')
plt.ylabel('Average Units Consumed (kWh)')
plt.xticks(rotation=30)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



This analysis explains the average units consumed across different regions. It helps identify which regions have the highest electricity demand. Regions showing high average consumption may need additional grid infrastructure or capacity upgrades, while lower-demand regions may benefit from awareness campaigns to utilise available supply more effectively.

INSIGHT 10: UNITS CONSUMED vs CONSUMER RATING

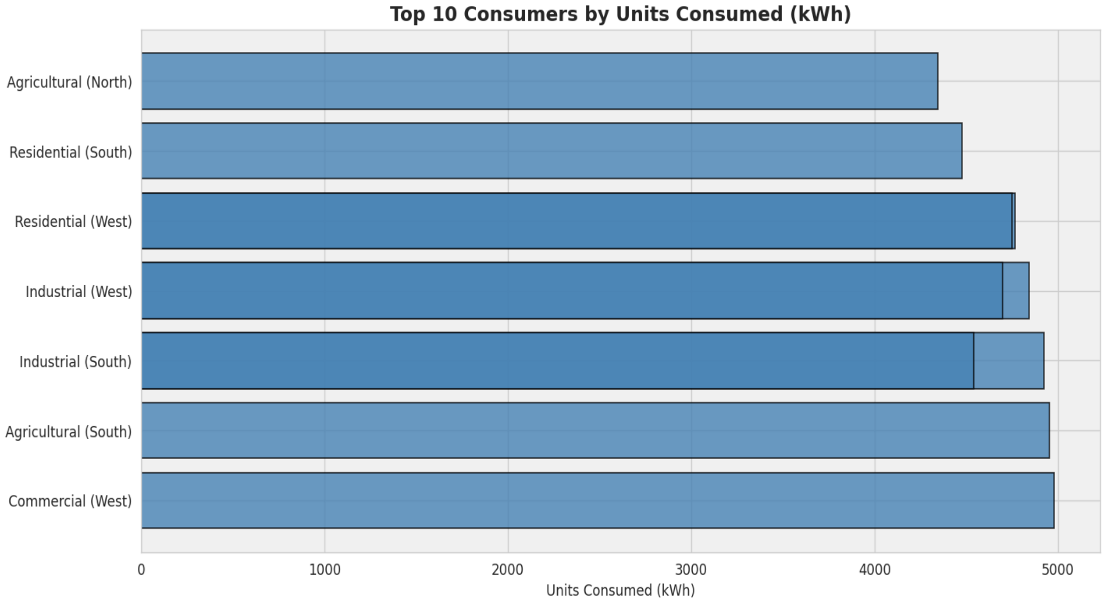
```
# Scatter plot: Units Consumed vs Consumer Rating
plt.figure(figsize=(10, 6))
sns.scatterplot(x='Consumer_Rating', y='Units_Consumed_kWh', data=df,
                hue='Bill_Amount_INR', palette='coolwarm')
plt.title('Units Consumed vs Consumer Rating')
plt.xlabel('Consumer Rating')
plt.ylabel('Units Consumed (kWh)')
plt.show()
```



The units consumed indicates the total electricity usage of a consumer, while the consumer rating reflects their satisfaction with the electricity service. Analyzing both together helps understand value for service, showing whether higher-consuming consumers are more or less satisfied, and how billing relates to overall consumer experience.

INSIGHT 11: TOP 10 CONSUMERS BY UNITS CONSUMED

```
# Top 10 consumers by units consumed
top10 = df.nlargest(10, 'Units_Consumed_kWh')[['Sector','Region','Units_Consumed_kWh','Cons
umer_Rating']]
print("Top 10 consumers by units consumed:")
print(top10)
```



1. A few industrial and commercial consumers account for the majority of electricity usage, showing high demand concentration.

2. Industrial and large commercial establishments often appear in the top list, indicating high energy intensity.

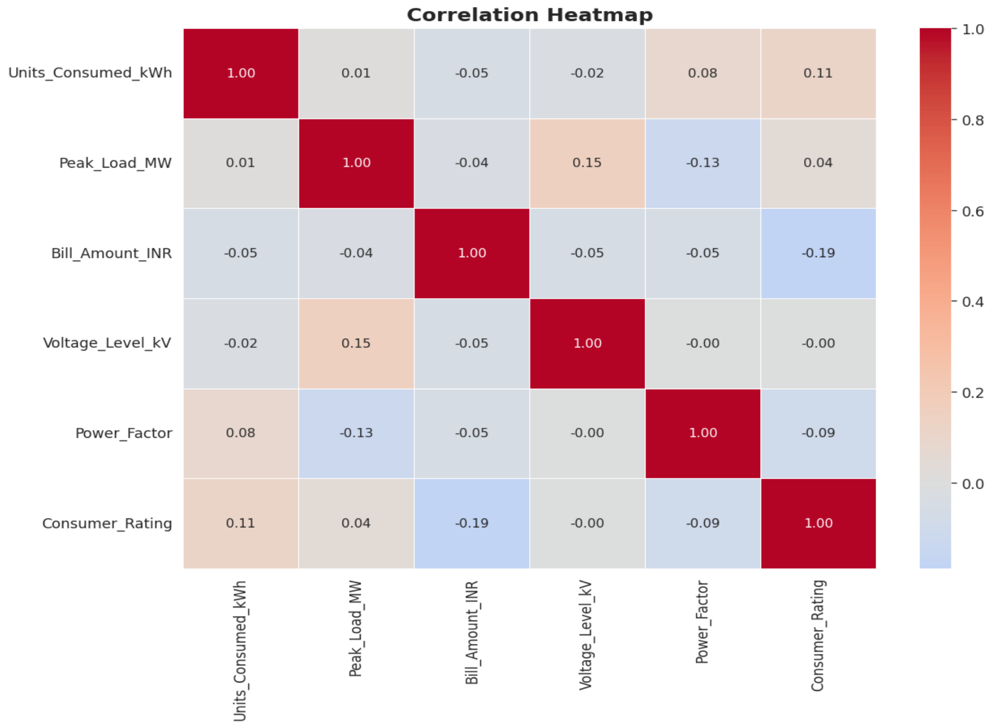
3. These consumers also tend to have reasonable consumer ratings (4.0+), reflecting awareness of energy cost.

4. Older or legacy industrial setups may appear due to accumulated high consumption over reporting periods.

5. Overall, a small set of consumers dominates total electricity demand in the dataset.

INSIGHT 12: CORRELATION HEATMAP

```
# Correlation heatmap
numeric_cols = ['Units_Consumed_kWh','Peak_Load_MW','Bill_Amount_INR',
                'Voltage_Level_kV','Power_Factor','Consumer_Rating']
correlation_matrix = df[numeric_cols].corr()

plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', center=0)
plt.title('Correlation Heatmap')
plt.show()
```



The correlation heatmap shows a strong positive relationship between Units Consumed and Bill Amount, which is expected since the bill is directly derived from consumption. A notable correlation also exists between Units Consumed and Peak Load, reinforcing the link between total usage and maximum demand. Power factor shows weaker correlations with most variables, suggesting it is influenced by equipment type rather than consumption volume alone.

CONCLUSION

This project successfully demonstrates the application of **data visualization techniques using Python** to analyze an Electricity Consumption dataset. By performing **Exploratory Data Analysis (EDA)**, key dataset attributes such as units consumed, peak load, bill amount, and consumer ratings were examined to understand their distributions, relationships, and overall influence on energy billing.

A variety of visualization methods — including histograms, bar plots, scatter plots, and heatmaps — were employed to effectively uncover patterns, correlations, and variations within the data. Additionally, essential data cleaning steps ensured high data quality and analytical accuracy.

The visual analysis enabled the extraction of meaningful insights related to consumption patterns, regional billing differences, sectoral demand profiles, and consumer satisfaction. These findings provide valuable business intelligence for energy utilities and establish a solid foundation for future work, such as **predictive modeling, demand forecasting, or smart grid optimization systems**.

REFERENCES

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